

Rongala Arun Siddardha

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in LLM inference performance and production GPU serving. At BlackBox AI, own the serving stack behind publicly benchmarked throughput leadership on Artificial Analysis: **#1 of 7 providers** on NVIDIA Nemotron 3 Ultra and **#2 of 17** on GLM-5.2 (max), where BlackBox is the fastest provider serving at BF16 precision. Deploy and optimize frontier MoE models up to **2.8 trillion parameters** on NVIDIA Hopper and Blackwell (H100-B300), spanning NVFP4/MXFP4/FP8 quantization, speculative-decoding draft training, distributed KV-cache, and multi-node fleets sustaining **51M TPM**. Also a core contributor (1,300+ commits) across the LLM API gateway, enterprise API, CLI and developer tooling, and built a multi-agent coding system scoring **94.7% pass@1 on Terminal-Bench 2.1**. Previously at Microsoft (Cloud + AI) developing RAG-based automation agents and ML-driven Azure migration tools. Over two years of professional experience; B.Tech in Artificial Intelligence from IIT Hyderabad (CGPA: 8.85).

TECHNICAL SKILLS

Languages: Python, Rust, C++, C#, Java, JavaScript, TypeScript, SQL

LLM Inference & Serving: vLLM, SGLang, TensorRT-LLM, Speculative Decoding (EAGLE3, DFlash, MTP), SpecForge, Speculators, NVIDIA ModelOpt, NVFP4 / MXFP4 / FP8 quantization, Distributed KV-Cache, Mooncake, LMCache, P/D Disaggregation, Tensor Parallelism, CUDA Graphs, Continuous Batching

GPU & Systems: NVIDIA H100 / H200 / B200 / B300 (Hopper & Blackwell), CUDA 12.8 / 13, Kernel Profiling & Fusion, MoE and Mamba/Linear-Attention Architectures, Intel TDX & Confidential Computing, Prometheus, Grafana

AI/ML: PyTorch, TensorFlow, Hugging Face, LangChain, FAISS, RAG, Multi-Agent Systems, MCP, Terminal-Bench, NLP, Jina, Crawl4AI

Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, .NET, GraphQL, Stripe, Langfuse

Cloud & Infra: Azure, AWS, Kubernetes, Docker, CI/CD, GitHub Actions, Caddy / NGINX, Terraform-style VM provisioning

Data: PostgreSQL, MongoDB, Redis, Pinecone, Qdrant, Git, WebRTC, VS Code Extension API

WORK EXPERIENCE

Senior Machine Learning Engineer

11/2025 – Present

BlackBox AI

Remote

- Own LLM inference performance and production serving; core contributor (1,300+ commits) across BlackBox AI's product suite: LLM API gateway, enterprise API, CLI, VS Code extension, AI coding platform, and remote code execution.
- **Public Benchmark Leadership (Artificial Analysis):** Own the single-stream serving configurations behind BlackBox AI's published throughput, ranked **#1 of 7 providers** on NVIDIA **Nemotron 3 Ultra** (436 tok/s) and **#2 of 17** on **GLM-5.2 (max)** (460 tok/s), where BlackBox is the **fastest provider serving at BF16** precision, ahead of all but one provider running 4-bit or 8-bit quantization.

- **Frontier-Scale Model Deployment:** Brought a **2.8-trillion-parameter** hybrid-MoE vision-language model (MXFP4-native, 1.5 TB across 96 shards, 93 layers of interleaved Mamba/linear-attention and MLA) into production on 8×B300 across both **SGLang** and **vLLM**, reaching **12/12 conformance** and 267 tok/s, with speculative decoding lifting decode **2.9×** (65 → 190 tok/s). Also deployed 550B–750B class MoE models (Nemotron-3-Ultra, GLM-5.2, Kimi K2.7-Code, Inkling) in NVFP4, FP8 and BF16.
- **Fleet Throughput & Capacity:** Built and tuned a multi-node 8×B300 serving fleet sustaining **51M TPM at 443 RPM** with 4.0s p50 TTFT and zero errors ($\sim 1.8\times$ the prior fleet) via 24 rank-endpoint sharding and cache-aware routing. Separately lifted per-GPU throughput from **600K → 2.3M TPM ($\sim 3.8\times$)** on the 8×B200 MiniMax fleet through batch-proxy tuning, P/D disaggregation and tensor-parallel re-sharding; ported the stack across H100/H200/B200/B300.
- **Speculative-Decoding Training (B200 / B300, CUDA 12.8):** Trained **EAGLE3** and **DFlash** draft models for **MiniMax M2.5/M2.7**, **Kimi K2.7-Code** and **GLM-5.2** on NVFP4 verifiers using SpecForge, achieving a **$\sim 40\%$ draft-token acceptance score**; converted checkpoints via Speculators for both **SGLang** and **vLLM** runtimes and validated end-to-end serving with EAGLE3 + FP8 KV-cache. Self-quantized GLM-5.2 BF16→NVFP4 with **ModelOpt** while preserving the MTP draft head that public checkpoints drop, restoring draft acceptance from 0 to 3.8 tokens.
- **Distributed KV-Cache & Runtime Stack:** Built shared KV-cache layer with **Mooncake** pooling and **LMCache** integration on top of vLLM/SGLang; profiled and patched vLLM scheduler / paged-attention and rejection-sampler paths and applied Torch-level optimizations (CUDA graphs, kernel fusion, NVFP4/FP8 dtype tuning) to raise context reuse and lower TTFT under heavy multi-tenant load.
- **Confidential Computing (Intel TDX + GPU-CC):** Designed and adversarially validated a trusted-execution enclave on 8×B300 that lets customers hold host root while remaining unable to read or modify proprietary code: fetch-to-tmpfs delivery, dm-verity integrity measurement chained to runtime registers, and working remote attestation verified against the Intel root of trust.
- **Self-Serve Deployment Platform:** Shipped a FastAPI service for one-call model deployment onto GPU cloud VMs (create / edit / delete, multi-replica, per-deployment auth), cutting provisioning from **~ 45 minutes to ~ 4** via pre-baked golden volumes, with cache-aware request routing, auto-provisioned load balancers, TLS issuance and per-deployment cost accounting.
- **Production Serving Ownership:** Run and maintain production inference endpoints end to end (reverse proxy → router → SGLang/vLLM) with Prometheus/Grafana monitoring, API-key hardening and automatic crash recovery; authored the conformance suites that gate every deploy and resolved customer-reported correctness defects in token accounting, generation limits and reasoning-channel handling.
- **Agentic Benchmarks:** Built a verify-then-escalate multi-agent coding system (primary solver, sandbox-executing critic, cross-family fallback) reaching **94.7% pass@1 on Terminal-Bench 2.1**, above the $\sim 89\%$ single-model ceiling; also built an LLM prompt-classifier auto-router with 99/100 routing accuracy at ~ 25 ms inference.
- **LLM API Gateway (AI Router), Top Contributor (240 commits):** Architected consistent-hash routing with prompt-cache-aware deployment affinity, adaptive thinking, per-request failover chains, auto-topup billing, multi-region load balancing with cache-aware distribution, and encrypted reasoning pipelines, serving millions of API calls across 20+ providers. Reduced TTFT via a 1,300-line TTFT tracking database, regional routing optimization and TPM/RPM token capping.
- **Enterprise API (37+ patches):** Shipped `/v1/responses` and `/v1/messages` APIs with multi-turn support, built interleaved thinking with cryptographic signature validation, tool-calling

sanitization, prompt caching with provider-specific breakpoints (OpenAI/Anthropic/Gemini), provider routing with latency/throughput sorting, streaming cost calculation, Codex ZDR, zero data retention, and reasoning effort controls. Patched upstream open-source proxy internals for Anthropic Messages and multi-provider gateway compatibility.

- **Blackbox CLI, #1 Contributor (381 commits):** Built multi-agent orchestration with parallel execution engine (PlannerAgent, ParallelAnalyzer, CoreToolScheduler), SQL database tool (1,289 lines), MCP server with auth and remote control (1,152 lines), encrypted model support, remote code execution, and Claude Code integration.
- **Developer Tooling (500+ commits):** Implemented the VS Code code-completion engine (Fill-in-Middle with Qwen, inline completions, repetition detection, prefix-aware context, 10+ models with GPU-accelerated E2E encryption); built the AI coding platform with Figma-to-code conversion, designer mode, multi-model support, billing and monitoring; and created **GithubConnector** as sole author, building semantic code search using FAISS and LangChain with multi-round progressive-deepening retrieval and Redis-based streaming.

Software Engineer, Cloud + AI

6/2024 – 11/2025

Microsoft

Hyderabad, India

- Developed AI-powered automation agents leveraging Retrieval-Augmented Generation (RAG) to process and summarize internal enterprise documents, improving operational efficiency by 30%.
- Engineered AI-driven workload migration solutions for Azure, integrating local LLMs and API-based models (OpenAI, Azure AI). Built ML models to analyze on-premises application dependencies using graph-based clustering and unsupervised learning, reducing customer decision-making phase for Azure migration by 30%.
- Designed and deployed scalable AI microservices on Azure Kubernetes Service (AKS) using event-driven architecture and CI/CD pipelines.

Software Engineer Intern, Cloud + AI

5/2023 – 7/2023

Microsoft

Hyderabad, India

- Reduced build times by **2 hours** by migrating from **CoreEXT** to **MSBuild** and resolved **1,800+ security vulnerabilities** by hardening build configurations across two repositories.

EDUCATION

Indian Institute of Technology, Hyderabad (IIT Hyderabad)

Hyderabad

B.Tech in Artificial Intelligence (CGPA: 8.85)

7/2024

PUBLICATIONS

PerfMon: Performance Monitoring of Host Network Stack. Ranjitha K., Ankit Sharma, Malsawmsanga Sailo, **Arun Siddardha**, Amrit Kumar, Praveen Tammana, Pravein Govindan Kannan, Priyanka Naik. *ACM Symposium on Cloud Computing (SoCC '25)*, Nov 2025. DOI: 10.1145/3772052.3772244 (IBM Research).

eBPF-based per-component latency monitoring across the host network stack with automatic baseline derivation and bottleneck detection for containerized microservices; evaluated on a Kubernetes bare-metal cluster with minimal overhead.

HONORS & AWARDS

JEE Mains 2020: AIR-19 (100 percentile) out of 0.87 million candidates

JEE Advanced 2020: AIR-610 out of 0.22 million candidates

KVPY Scholar: AIR-281 among 0.15 million candidates